Product Overview: A Digital Ecosystem for Youth Mentorship

This application is a comprehensive, secure, and AI-enhanced case management platform designed to bridge the gap between social workers and the children they mentor. It streamlines communication, centralizes case data, and leverages AI to provide actionable insights, enabling social workers to focus more on intervention and less on administrative tasks.

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1. User Onboarding & Authentication

Objective: To provide a secure and role-appropriate entry point for all users.

- **Landing Page (**main.tsx**):** The primary entry point for all users. It acts as a central hub, featuring intuitive navigation links to the login and registration pages for both social workers and children.
- **Registration (**register.tsx**):** A guided process where new users select their role (Social Worker or Child).
    - **Child Accounts:** Registration for children is restricted. Accounts for minors must be created by an authorized social worker.
    - **Social Worker Registration:** Requires standard profile details (Full Name, Username, Email, Password) to set up a secure account.
- **Login (**login.tsx**):** A streamlined, role-based login screen. Users select their role and enter their credentials. Upon successful authentication, the system dynamically routes them to their respective, personalized dashboards (Social Worker Home or Child Home).

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2. Social Worker Dashboard & Case Management

Objective: To provide social workers with a centralized, efficient, and data-rich environment to manage their caseload and collaborate with the children they support.

- **Home Dashboard (**home.tsx**):** The landing page after login. It provides a high-level overview, displaying a list of the 10 most recently interacted children. A global search bar allows for quick access to any child's profile. Selecting a child from this list updates their status to the "most recently interacted" for quick follow-up.
- **Active Cases (**activeCases.tsx**):** This is the administrative and clinical command center for a social worker.
    - **Digital Clipboard:** Centralizes all critical information for a case, including AI-summarized insights from the child's chatbot interactions and daily check-ins.
    - **Risk Assessment:** Features an integrated CANS (Child and Adolescent Needs and Strengths) assessment section where workers can rate risks, log detailed case notes, and record follow-up actions.
    - **Intelligent Interventions:**

        - **Escalation/De-escalation:** Allows workers to adjust a child's risk level based on their professional judgment and AI recommendations.
        - **Case Closure:** If a child is ready to transition out of the program, a social worker can securely remove the profile, a process that requires email verification for security and audit purposes.
    - **Quick Actions:** Direct buttons allow the worker to:
        - **"Start Meetup Session":** Initiates a voice-recorded session with AI running in the background. After the session ends, the AI automatically processes the conversation and populates relevant fields in the active case and documentation.
        - **"Visit Catalog":** Provides direct access to the child's detailed documentation in the youth catalog.
- **Calendar & Collaboration (**calendar.tsx**):** A robust scheduling tool for planning events and check-ins.
    - **Event Management:** Users can create, manage, and delete events. Event creation includes a "collaboration" feature, allowing social workers to invite colleagues and children. **All events must include at least one social worker and one child to ensure coordinated care.**
    - **Permissions:** Only the social worker who creates an event has the authority to delete it. All invited participants must accept the invite for it to be added to their calendars.
- **Youth Catalog (**youthCatalog.tsx**):** A comprehensive, searchable directory for all children in the system.
    - **Profile Creation:** This is the secure hub where social workers can create accounts for children, requiring essential details like Full Name, Date of Birth, Username, Email, and Password.
    - **Detailed Profiles:** Clicking a child's profile reveals a detailed documentation page containing key demographic and background information (e.g., NRIC, School, Hobbies).
    - **Additional Documentation:** This section provides an area for social workers to add extra, unstructured notes beyond the structured fields in activeCases.
- **Messaging & AI Integration (**messages.tsx **- Social Worker View):** A secure, WhatsApp-style messaging platform for real-time communication between social workers and children.
    - **Organized Chat:** Each child has a dedicated and separate chat window.
    - **AI-Driven Insights:** Information gathered from conversations is automatically captured and sent to activeCases.tsx for that specific child, ensuring no actionable information is lost.

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3. Child Dashboard & Engagement Tools

Objective: To empower children with a safe, engaging, and supportive digital space for self-expression, communication, and growth.

- **Home Dashboard (**home.tsx**):** The child's landing page. It displays a list of their 10 most recently interacted social workers, with a search bar for finding specific contacts. Clicking a social worker's name marks them as the most recent point of contact.
- **Wellness Check-Ins (**checkIns.tsx**):** A daily emotional and well-being check-in tool.
    - **Quick Mood Rating:** Children can select an emoji ranging from 😄 (Very Positive) to 😢 (Very Negative) to express their current feelings.
    - **Reflective Journaling:** Prompts encourage children to share positive and negative events.
    - **Wellness Questions:** Includes 5 additional well-being questions to provide a holistic view of the child's state. All data is automatically and securely transmitted to their social worker's activeCases.tsx.
- **Messaging (**messages.tsx **- Child View):** A simple, intuitive messaging interface. Children can view their 10 most recent social worker contacts and initiate a conversation. This secure chat is the child's primary channel for direct communication with their mentor.
- **Calendar (**calendar.tsx**):** A view-only calendar for children, displaying all confirmed events. For each event, the child can see the names of all invited participants (other children and social workers), fostering transparency and clarity.
- **AI Chatbot (**chatbot.tsx**):** An intelligent, therapeutic chatbot for after-hours support.
    - **Availability:** Designed as a support resource during off-peak hours (6 PM – 7 AM SGT).
    - **Conversation Management:** Children can create, manage, and delete multiple chat threads, providing a flexible and private interaction space.
    - **Privacy & AI Learning:** All conversations are summarized by the AI and securely transmitted to the child's social worker to inform case management without exposing the raw chat log, ensuring privacy and actionable insights.

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4. Brand Identity & Design Language

The application is built with a calming, approachable, and professional visual identity. It uses a **"Deep Ocean Blue"** primary color to convey trust and stability, complemented by **"Growth Green"** to symbolize progress and well-being. The typography is modern and human-centric, with **Fredoka** for welcoming headings and **Plus Jakarta Sans** for clear, readable UI.

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5. Technology Stack & Architecture

- **Frontend:** React with TypeScript for a robust, maintainable, and scalable single-page application.
- **Backend:** Node.js (implied) with Express to handle API requests, business logic, and authentication.
- **Authentication & Authorization:** JWT (JSON Web Tokens) for secure, stateless authentication, with role-based access control for all features.

Conclusion

This platform is not just a database; it is a dynamic, collaborative ecosystem that empowers social workers with data-driven insights and provides children with a supportive digital environment. It is designed to enhance efficiency, improve communication, and ultimately, deliver better outcomes for the youth we serve.